

PW1: Object-relational databases

Domain description: HOSPITAL

Your domain is: the hospital.

The hospital's primary function is to provide comprehensive healthcare services, including diagnosing, treating, and monitoring patients. The system supports the management of medications, prescriptions, healthcare providers, and patient records, ensuring smooth and secure care processes.

Doctors and pharmacists handle the administration and monitoring of medications. Each medication has a unique identifier, a name, and a description, and can be detailed further as needed. Medications may consist of various ingredients, each with its own ID and name, and are categorized by type, such as oral medications (e.g., tablets or syrups, with properties like flavor or color) and injectable medications (e.g., specifying needle size or injection type).

Patients are registered with unique IDs and personal data that link them to medical records and prescriptions. Prescriptions include essential details like medication, dosage, and administration method, as well as doctor and patient identifiers. Doctors can create, update, and complete prescriptions based on treatment progress, ensuring that each patient's records reflect the current care plan.

The hospital's medical staff comprises doctors, nurses, and pharmacists. Doctors, each identified by a unique license number and salary, are responsible for prescribing medications and monitoring treatments, while nurses and pharmacists prepare and administer medications. Staff interact with the database through hospital-approved software that upholds patient confidentiality by restricting access to sensitive information according to hospital policies.

The hospital database supports the workflow needed to keep patient and treatment information up-to-date, from tracking prescription statuses to recording details on medication administration. Only authorized medical staff can access and manage this information, ensuring data accuracy, patient privacy, and streamlined care delivery across the healthcare team.